Assignment No 8

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
kd2_98837_sakshi> select * from orders
    -> where cnum=(select cnum from customers where cname='Cisneros');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.00 sec)
```

7. Extract all the orders of Motika.

```
kd2_98837_sakshi> select * from orders
    -> where snum=(select snum from salespeople where sname='Motika');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+------+---------+------------+------+------+
1 row in set (0.00 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
kd2_98837_sakshi> select * from orders
    -> where amt>ANY(select cnum from customers where city='london');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.00 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in London.

```
kd2_98837_sakshi> select * from orders
    -> where amt>ANY(select amt from orders o
    -> inner join customers c on o.cnum=c.cnum
    -> where city='LONDON');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
6 rows in set (0.00 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'.
(Hint: find all customers of serres salesperson).

```
kd2_98837_sakshi> select * from customers where rating>=ANY
    -> (select rating from customers c
    -> inner join salespeople s on s.snum=c.snum
    -> where s.sname='serres');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

11. Select customers who have a greater rating than any other customer in Rome.

```
kd2_98837_sakshi> select * from customers where rating > ANY (select rating from customers where city='Rome');
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

12. Select all orders that had amounts that were greater that at least one of the orders from '1990-10-04'.

```
kd2_98837_sakshi> select * from orders
    -> where amt>ANY(select amt from orders where odate>='1990-10-04');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
kd2_98837_sakshi> select * from orders
    -> where amt < ANY(select amt from orders o
    -> inner join salespeople s on o.snum=s.snum
    -> where s.city='San Jose');
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
8 rows in set (0.00 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
kd2_98837_sakshi> select o.amt,c.cname,c.rating from orders o
    -> inner join customers c on c.cnum=o.cnum
    -> where o.amt>(select AVG(amt)from orders);
+---------+---------+--------+
| amt     | cname   | rating |
+---------+---------+--------+
| 5160.45 | Liu     |    200 |
| 9891.88 | Clemens |    100 |
| 4723.00 | Clemens |    100 |
+---------+---------+--------+
3 rows in set (0.00 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
kd2_98837_sakshi> select * from customers
    -> where rating>ALL(select rating from customers where city='london'
);
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2002 | Giovanni | Rome     |    200 | 1003 |
| 2003 | Liu      | San Jose |    200 | 1002 |
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
4 rows in set (0.00 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
kd2_98837_sakshi> select sname,snum,city from salespeople where city != All
    -> (select city from customers where salespeople.snum=customers.snum);
+---------+------+-----------+
| sname   | snum | city      |
+---------+------+-----------+
| Motika  | 1004 | London    |
| Rifkin  | 1007 | Barcelona |
| Axelrod | 1003 | New York  |
+---------+------+-----------+
3 rows in set (0.00 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
kd2_98837_sakshi> select snum,sname,city from salespeople s where (select count(*) from customers c
    -> where c.snum=s.snum)>1;
+------+--------+----------+
| snum | sname  | city     |
+------+--------+----------+
| 1001 | Peel   | London   |
| 1002 | Serres | San Jose |
+------+--------+----------+
2 rows in set (0.00 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
kd2_98837_sakshi> select snum,sname,city,(select count(*) from customers c
    ->     where c.snum=s.snum) as customer_count
    ->     from salespeople s where (select count(*) from customers c
    ->     where c.snum=s.snum)>1;
+------+--------+----------+----------------+
| snum | sname  | city     | customer_count |
+------+--------+----------+----------------+
| 1001 | Peel   | London   |              2 |
| 1002 | Serres | San Jose |              2 |
+------+--------+----------+----------------+
2 rows in set (0.00 sec)
```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
kd2_98837_sakshi> select snum,SUM(amt) from orders
    -> group by snum
    -> having sum(amt)>(select max(amt) from orders);
+------+----------+
| snum | SUM(amt) |
+------+----------+
| 1001 | 15382.07 |
+------+----------+
1 row in set (0.00 sec)
```